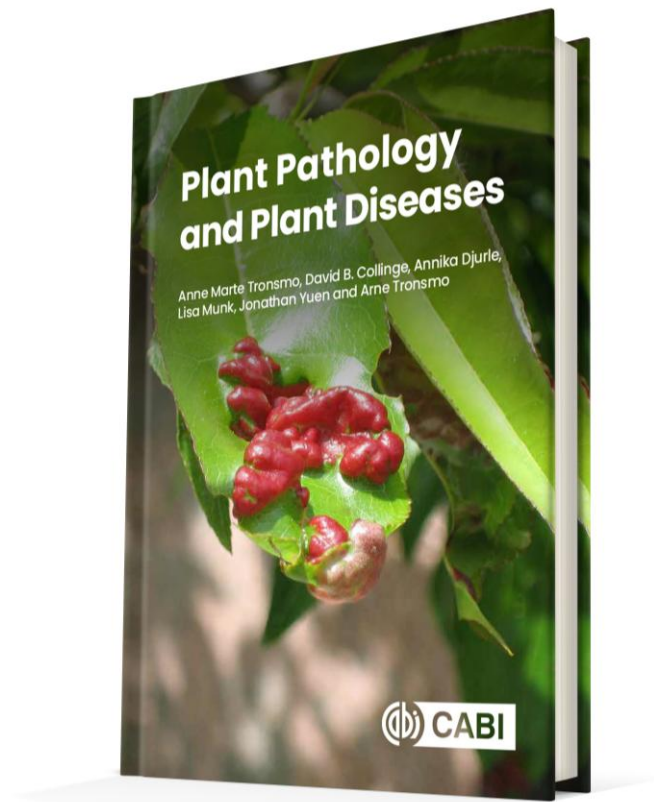




Plant Pathology and Plant Diseases

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Lisa Munk, Jonathan Yuen and Arne Tronsmo 2020

Fig. 17.1 Screening procedure conducted in soil. The potential antagonists are introduced in soil at sowing time in a mixture with the pathogen *F. oxysporum* f.sp. *lini*. (© C. Alabouvette.)



Fig. 17.2 Four mechanisms for pathogen control by endophytes.
(© Latz, et al (2018) Plant Ecology & Diversity 11, 555–567)

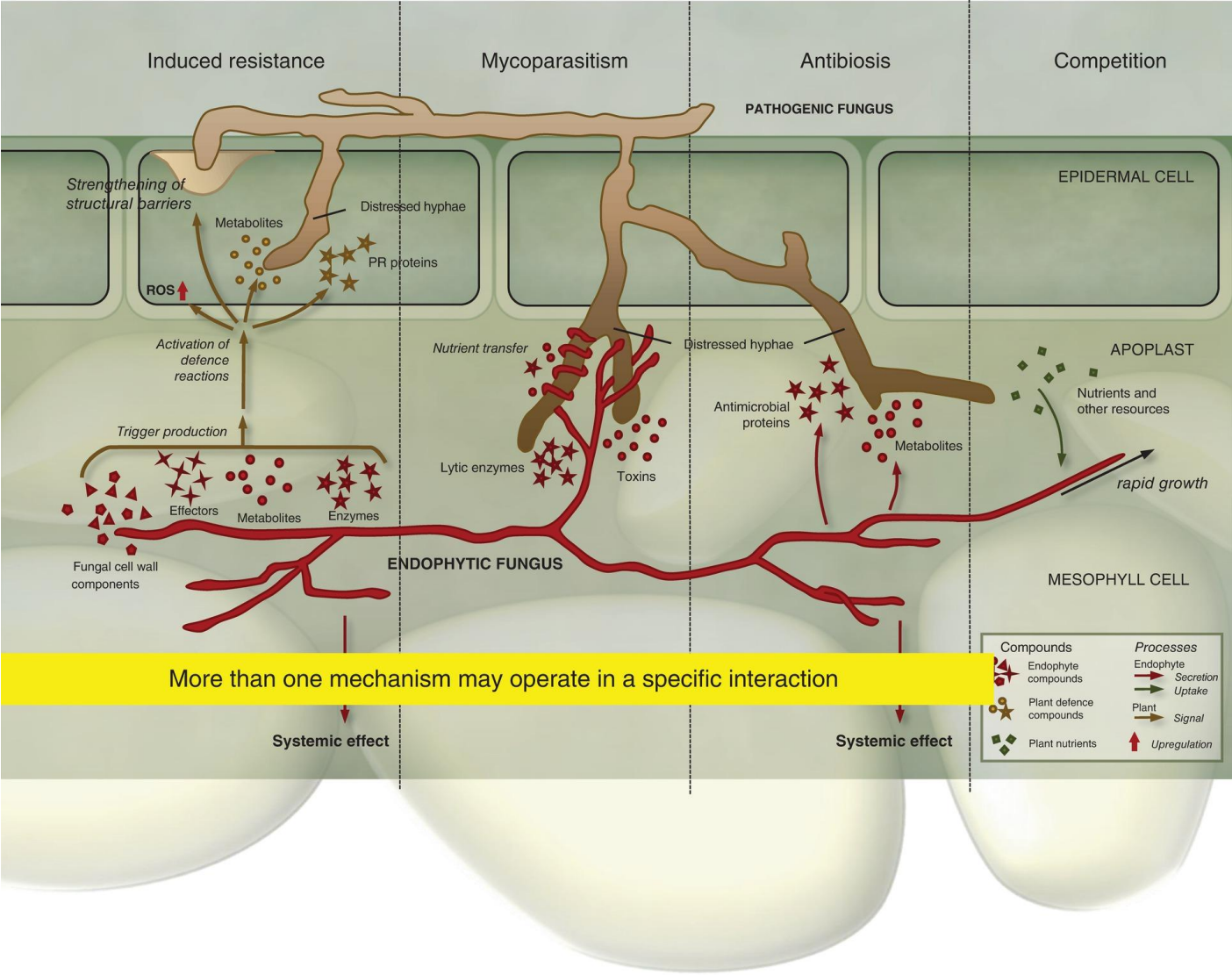


Fig. 17.3 Antibiosis by *Pseudomonas* sp. versus *Pythium ultimum*. One strain of *Pseudomonas* inhibits the growth of *Pythium*, the other has no effect, the pathogen grows over the bacteria. (© C. Alabouvette.)

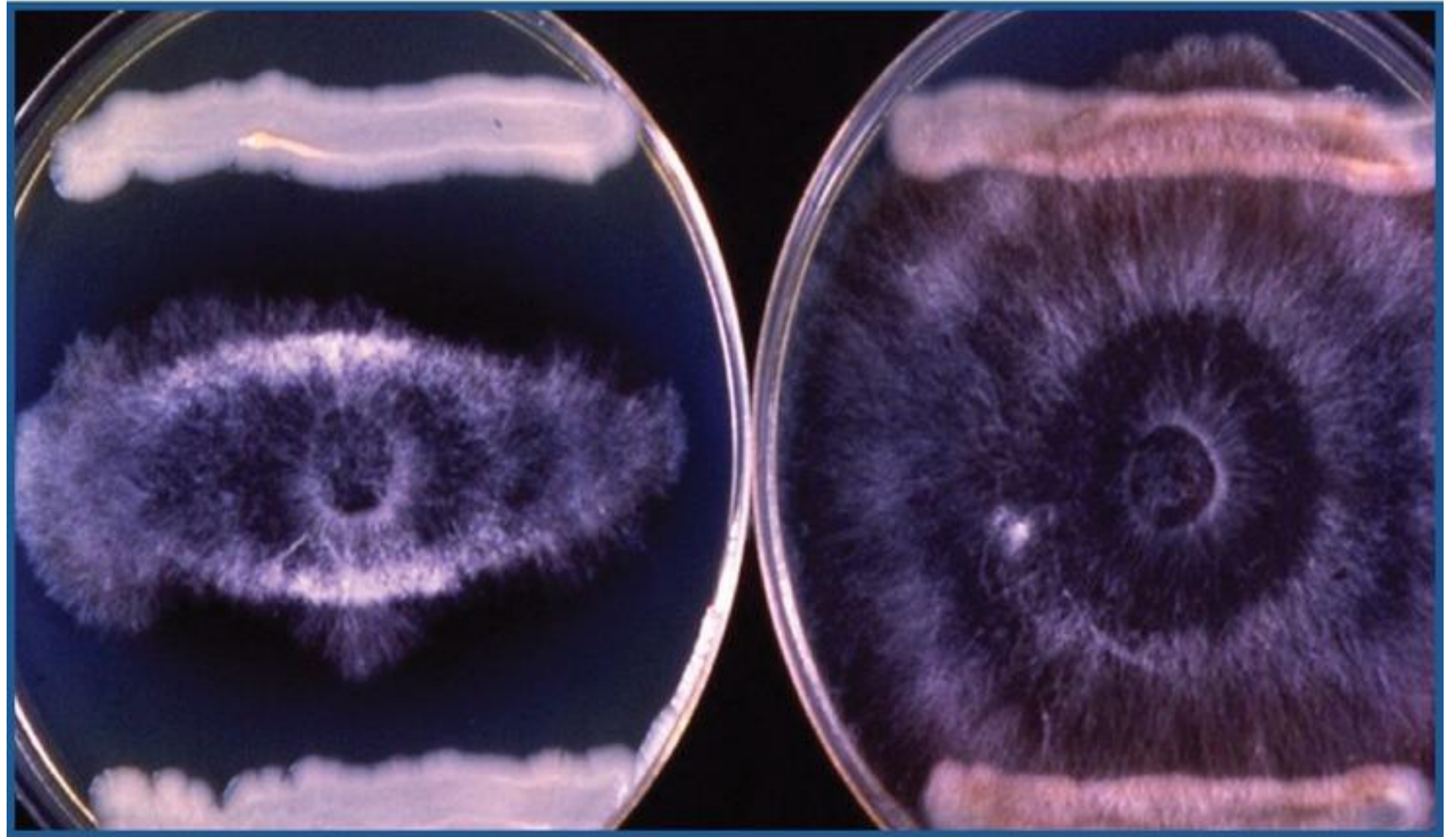


Fig. 17.4A Hyperparasitism.

(A) Thin hyphae of *Trichoderma*, coiling around and penetrating into large hyphae of *Rhizoctonia solani*.
(© C. Alabouvette)



Fig. 17.4B Hyperparasitism.

(B) Powdery mildew (*Sphaerotheca fuliginea*) on cucumber. Leaf to the left treated with the biocontrol fungus *Tilletiopsis minor*, leaf to the right untreated control. (© J. S. Hansen.)



Fig. 17.4C Hyperparasitism.

(C) Scanning micrograph of the cucumber leaf. *T. minor* is parasitizing the cucumber powdery mildew hyphae and conidiophore. (© J. S. Hansen.)

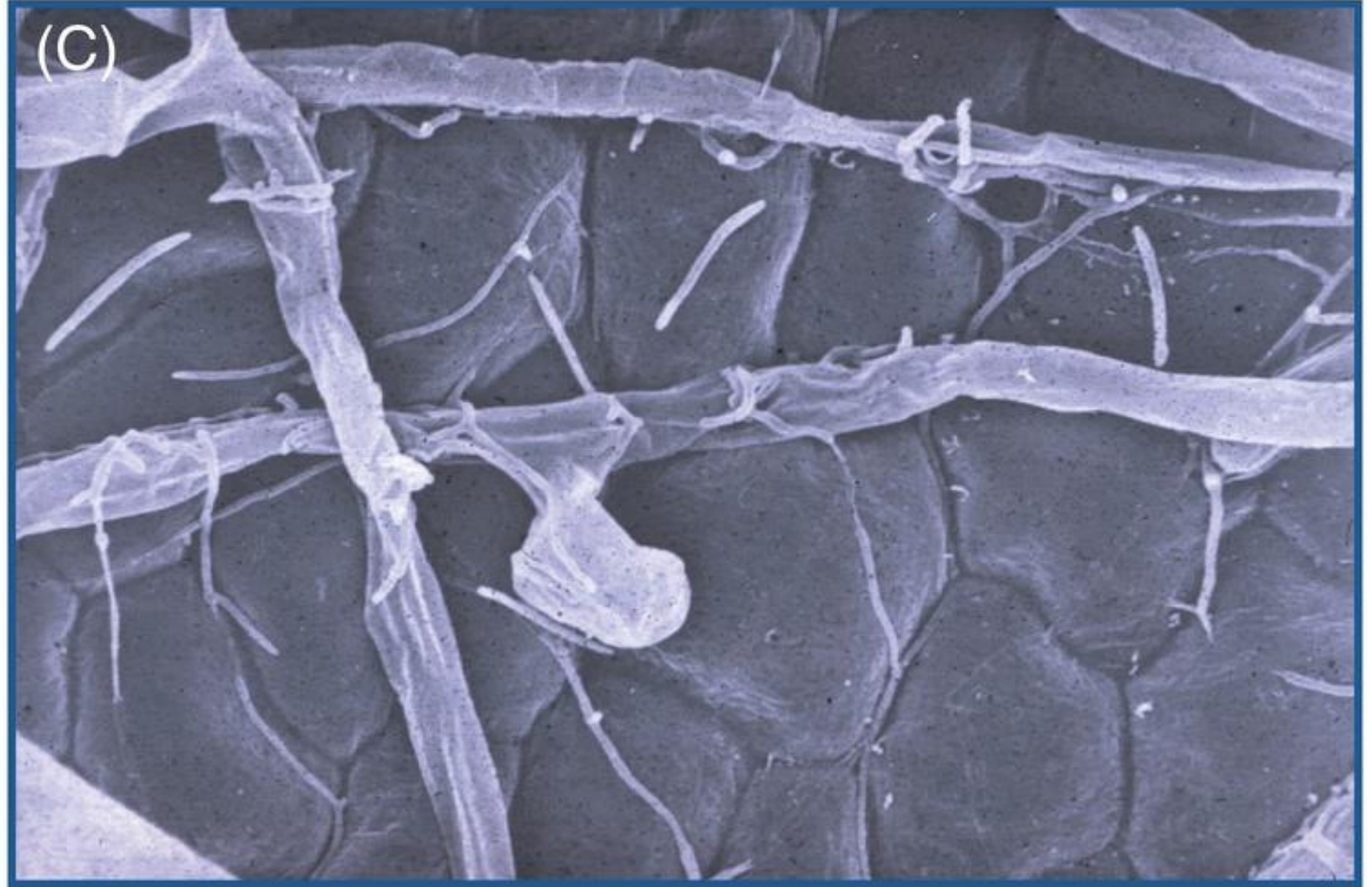


Fig. 17.5 The effect of priming: Chitinase 3 is over-expressed in tomato root first inoculated with Fo47 then by the pathogenic strain. (© C. Alabouvette.)

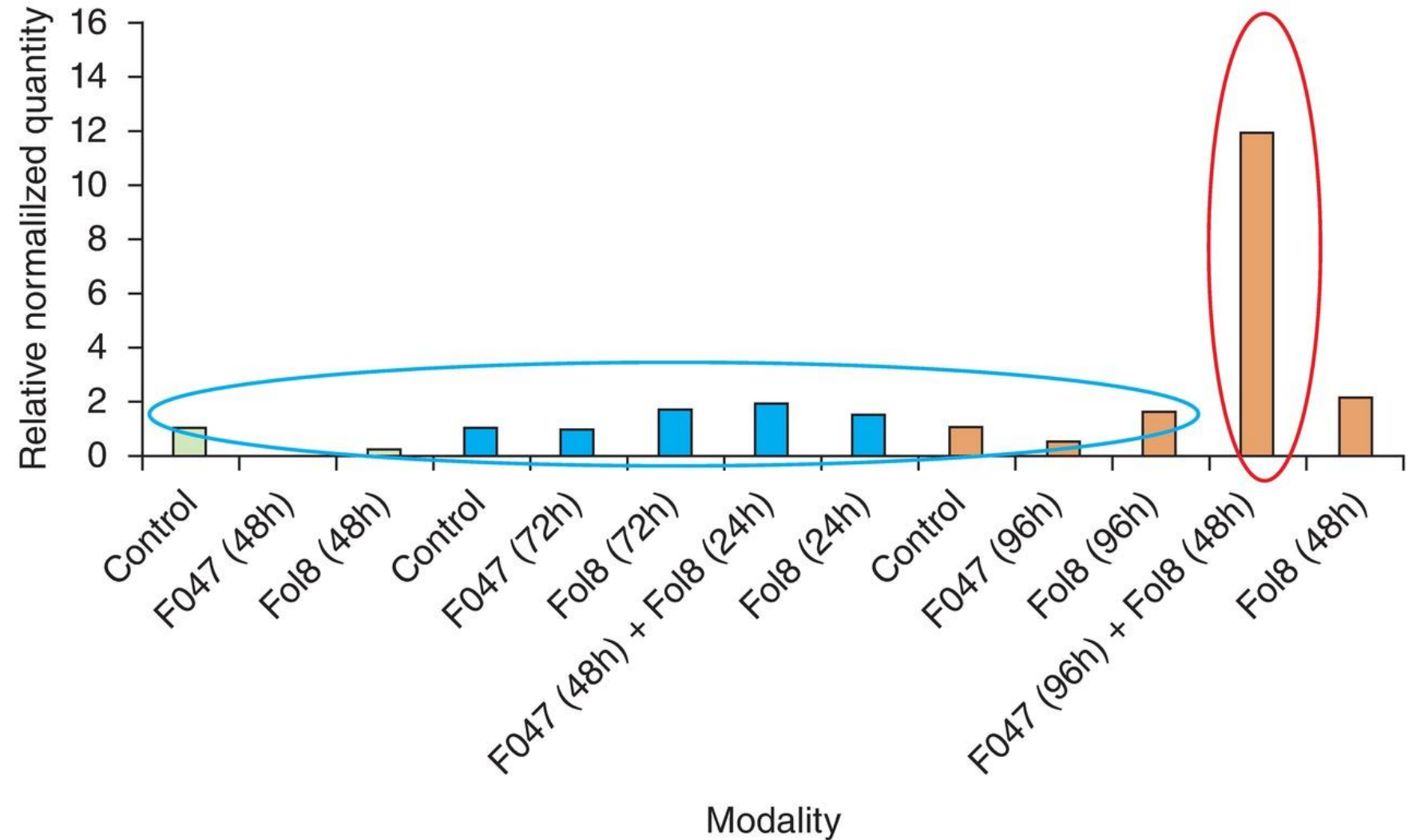


Fig. 17.6 The effect on growth of non-inoculated control plants (NI) and plants inoculated by two different strains (L and B) and the mixture of the two strains. Using two strains was not beneficial. (© C. Alabouvette.)



Fig. 17.7 Role of the soil microflora in the mechanisms of fusarium wilt suppressive soil. % of wilted plants in suppressive soil (SS), steamed SS, steamed SS with a strain of nonpathogenic *F. oxysporum*, (steamed +Fo), a consortium of bacteria and fungi isolated from the SS (steamed SS+Mic), and the consortium of microorganisms + the strain of *F. oxysporum* (steamed SS+Mic+Fo).
(© C. Alabouvette.)

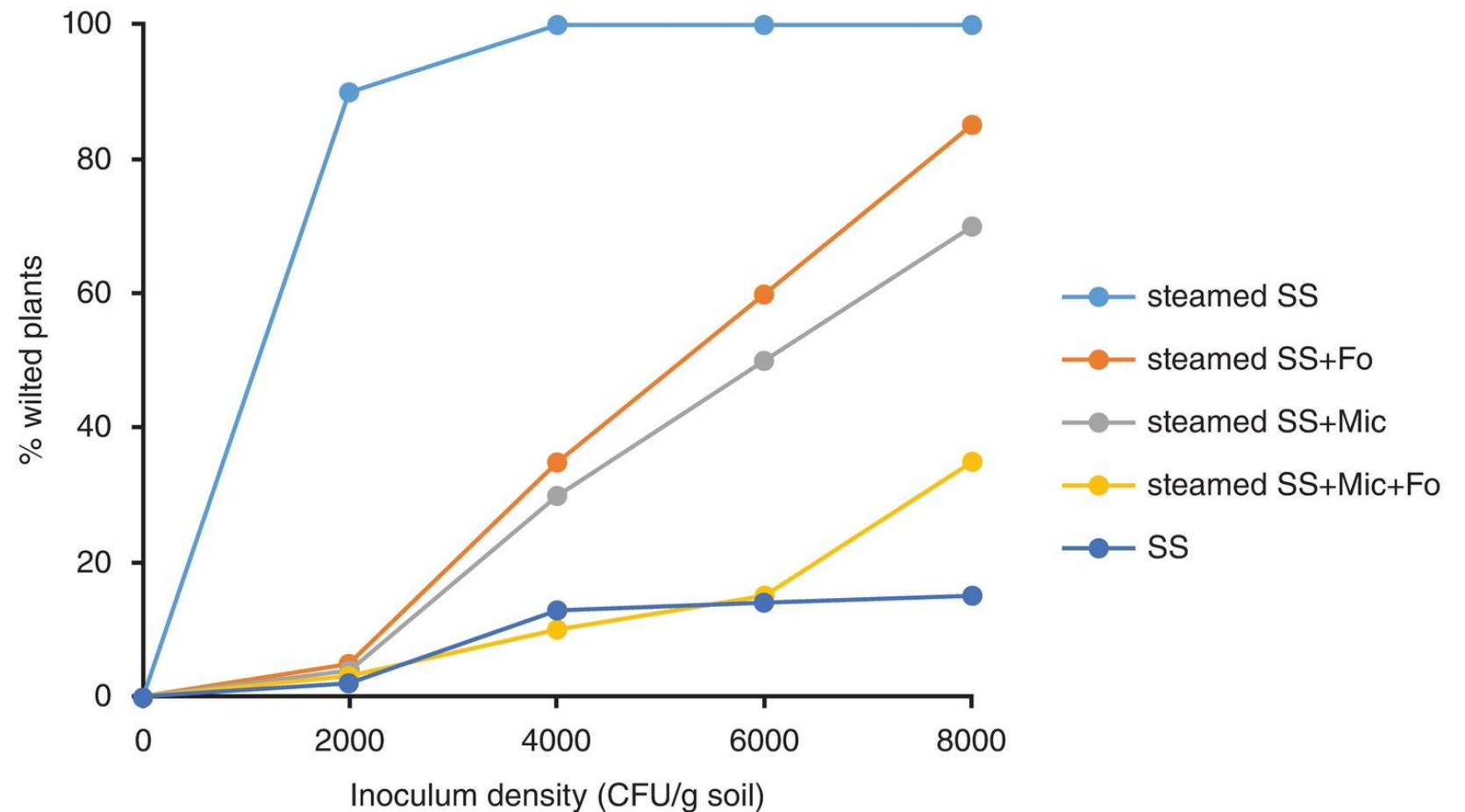


Fig. 17.8 Hyperparasitism. (A) Green *Trichoderma* sp. sporulating after infecting the black sclerotia of *Sclerotinia sclerotiorum*. (B) A fluorescent micrograph of colonization of *Sclerotium rolfsii* sclerotia by GFP transformed *Trichoderma virens* I10 3.4.10 GFP A. (A: © Daohong Jiang; B: © Sabrina Sarrocco.)

